Review Questions for Lecture 11

1. The fraction of people in a population who have a certain disease is 0.01. A diagnostic test is available to test for the disease, but the test isn't perfect. For a healthy person, the chance of being falsely diagnosed as having the disease is 0.05. For someone with the disease, the chance of being falsely diagnosed as healthy is 0.2. Suppose that a doctor performs the test on a person selected at random from the population.
   a. What are the prior odds for having the disease?
   b. What is the probability that the test shows a positive result (meaning the person is diagnosed as diseased, perhaps correctly, perhaps not)?
   c. What are the data odds for having the disease given a positive test result?
   d. What is the probability that the person is one who has the disease but is diagnosed as healthy?
   e. What is the probability that the person is correctly diagnosed and is healthy?
   f. Suppose the test shows a positive result. What is the probability that the person tested actually has the disease?
2. In jar A there are 7 red marbles and 3 green marbles. In jar B there are 2 red marbles and 8 green marbles. In jar C there are 5 red marbles and 5 green marbles. You flip a fair coin twice. If it lands heads both times, you pick a marble at random out of jar A. If it lands heads exactly one time, you pick a marble at random out of jar B. If it lands tails both times, you pick a marble at random out of jar C.
   a. What is the chance of picking jar A?
   b. What is the chance of getting a red marble?
   c. If you end up with a red marble, what is the chance it came from jar A?

3. In a math class 60% of the students are juniors and the rest are seniors. Of the juniors, 45% are math majors. Of the seniors, 85% are math majors. You pick one student at random.

   a. What is the chance that the student is a senior math major?
   b. Given that the student is a math major, what is the chance that the student is a senior?
   c. What is the chance that the student is a senior or a math major?

4. A company makes computer chips. Minor defects in the materials (silicon, etc.) are common, and the probability that any one chip is defective is 25%. Therefore, the company tests each chip to make sure it is good before it sends it to market. The test that the company uses is not perfect – even a bad chip will pass the test 10% of the time, and a good chip will fail 5% of the time. In the long run, what fraction of the computer chips that the company sells are defective?

5. (Note: This question is easier to answer after completing the next lecture.) A boy has an empty box. He flips a coin three times. Each time the coin lands heads, he puts a red marble in the box. Each time the coin lands tails, he puts a green marble in the box. He then reaches in the box and pulls out one of the three marbles at random.

a. What is the chance that the boy puts exactly two green marbles in the box?
b. What is the chance the boy puts at least one green marble in the box?
c. What is the chance the boy pulls out a green marble from the box?
d. Given that the coin lands heads exactly once, what is the chance that the boy pulls out a green marble from the box?
e. Given that the boy pulls out a green marble from the box, what is the chance that the coin landed heads exactly once?
f. Given that the boy pulls out a green marble from the box, what is the chance that the coin landed heads at least once?
g. Given that the boy pulls out a green marble from the box, what is the chance that the coin landed tails at least once?

6. Provide a long-run frequency interpretation for the probability calculated in part c) of question 2. Provide a long-run frequency interpretation for the probabilities calculated in parts f) and g) of question 5.

7. Suppose that a certain type of cancer is usually benign – in only one out of ten thousand cases does the cancer metastasize, leading to death. Suppose there is a diagnostic test available that can distinguish the benign cancers from the dangerous ones. Assume that the error rate of the test is 1%, meaning both the false positive and false negative error rates are 1%. Without doing any calculations, choose one of the following options, and then provide an intuitive explanation for your answer. If the doctor gives the test to a randomly selected individual with the cancer, and the results come back that the cancer is dangerous, the chance that the cancer actually is dangerous is:

a. about equal to 99%,
b. quite a bit higher than even 99%, or
c. quite a bit lower than 99%.

8. This question continues question 7.

a. What are the prior odds that the cancer is dangerous?
b. What are the data odds that the cancer is dangerous given a positive test?
c. What are the posterior odds that the cancer is dangerous given a positive test?
d. What is the probability that the cancer is dangerous given a positive test?
e. What would the data odds need to be for positive test to conclude that the there is a 50% chance for the cancer to be dangerous.

9. This question continues questions 7 and 8. Now suppose that there is a surgery that can remove the cancer, and that 95% of the time the surgery is safe and effective, but 5% of the time complications arise in surgery that result in the death of the patient. Suppose a randomly selected individual with the cancer is given the test, and the results come back that the cancer is dangerous. Given the test results, what is the chance the patient will die

a. undergoing surgery, or
b. doing nothing?

Comment briefly, noting especially that the test is accurate 99% of the time, and the surgery is safe and effective 95% of the time.

10. This question continues questions 7, 8, and 9. If you were a doctor, would you recommend your patient to have the diagnostic test? Explain your reasoning. (There are ethical issues involved, and no “right” answer. Just make sure that your answer makes the issues clear.)

11. Provide a long-run frequency interpretation for the probability in question 8.